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The Impact of Web 3.0 On E-Learning

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***Abstract*— Though The term Web 3.0 has become a subject of interest and debate since late 2006 to till date and not yet confirmed, there are Views of different pioneers on the evolution of Web 3.0.**

This paper introduces the basic concept of Web 3.0 and its main characteristics; semantic web, 3D web, social web, intelligent web. Moreover, the paper will outline the key semantic web technologies that allows to build applications and solutions that were previously impossible. Finally, the paper shall look in depth at Web 3.0 tools for e-learning.

Keywords— ***Web 3.0; Semantic Web; Educational Technology; Online Learnin; 3D Learning Environment; E-learning.***

I. INTRODUCTION

"The World Wide Web (WWW) project aims to allow links to be made to any information anywhere"[1]. The world wide web has developed from a text-based static pages that it is the first version of the web "web 1.0" -where the learners could only read the educational content shared over the internet- towards a Read/Write web "web 2.0" that enables the learners to interact and manipulate the educational content and finally into this new version a Read/Write/Execute web "web 3.0" that enables to combine and integrate Web content and services to improve the end-user experience. Therefore, Web 3.0 is an evolving extension of www. It is 3D, media centric, social, intelligent and semantic.

II. WEB 3.0 BASIC CONCEPT

Though The term Web 3.0 has become a subject of interest and debate since late 2006 to till date and not yet confirmed, there are Views of different pioneers on the evolution of Web 3.0.

The term Web 3.0 was coined in an article in The New York Times by John Markoff in 2006 "commercial interest in Web 3.0 - or the 'Semantic Web,' for the idea of adding meaning - is only now emerging.", and first appeared significantly in a Blog article "Critical of Web 2.0 and associated technologies such as Ajax"[2].

"People keep asking what Web 3.0 is. I think maybe when you've got an overlay of scalable vector graphics - everything rippling and folding and looking misty - on Web 2.0 and access to a semantic Web integrated across a huge space of data, you'll have access to an unbelievable data resource" [3].

" There has been a tendency to give this new evolutionary stage of the Web its own name: Web 3.0. We can thus essentially view Web 3.0 as Semantic Web technologies integrated into, or powering, large-scale Web applications" [4]. "Web 3.0, where you're actually generating new knowledge, the computer is generating new information, rather than perhaps with Web 2.0 where the humans are generating the actions" [5].

" While the creation of blogs, wikis, podcasts and video clips supports the so called Web 2.0, the materialization of the new Web 3.0 is apparent in Second Life, Divvio, Joost and VRML/X3D worlds and seems to be marked by this mix of humanlike avatars, intelligent agents and rich multimedia features that live happily within interactive 3D environments"[6].

Finally, we consider what Google's CEO, Eric Schmidt **[7]** stated: "Web3.0 as a series of combined applications. The core software technology of Web3.0 is artificial intelligence, which can intelligently learn and understand semantics. Therefore, the application of Web3.0 technology enables the Internet to be more personalized, accurate and intelligent."

III. CHARACTERISTICS OF WEB 3.0

A. Semantic Web

Tim Berners-Lee, founder of the Web, presents his vision of the Semantic Web as an extension of the current web where content is machine-readable and well defined enabling new functionality that will extends beyond the current capabilities of the Web[8]. According to him: "The Semantic Web will bring structure to the meaningful content of Web pages, creating an environment where software agents roaming from page to page can readily carry out sophisticated tasks for users" [8]. It is the transformation of the Web from a linked document repository into a distributed knowledge base and application

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platform. This platform will allow for the vast range of content and services to be effectively exploited [9].

W3C provides the following definition of the Semantic Web: The Semantic Web is a Web of data. There is a lot of data we all use every day, and it is not part of the Web. For example, I can see my bank statements on the web, and my photographs, and I can see my appointments in a calendar. But can I see my photos in a calendar to see what I was doing when I took them? Can I see bank statement lines in a calendar? Why not? Because we don't have a web of data. Because data is controlled by applications, and each application keeps it to itself [10].

B. 3D Web

Web 3.0 is a popular 3D internet application. It is a computer-based simulated 3D environment intended for its users to inhabit and interact via avatars [11]. High speed Internet, quicker processing speeds, higher screen resolutions, 3D gaming technology and augmented reality will transform the Web browsing into a 3D experience, where you actually move through the virtual corridors of the Web, as a virtual avatar of your real self [12].

- 3D global applications: 3D applications like Google Earth (2004, which is more of an Internet application than a Web application since has its own browser application), Virtual Earth (2004, a full Web application that runs inside the browser.
- 3D online Games: Massive Multi-player Online Role Playing Games (MMORPG) like World of Warcraft with more than 8 million online gamers.
- 3D Virtual worlds & Avatars: a 3D virtual world is a mix of 3D gaming technology, augmented reality, simulated environment powered with Internet technology where users interact through movable avatars [13].
- 3D Wikis: A 3D Wiki is a virtual 3D encyclopedia. An examples of this kind of technology can be found on software like Copernicus-3D Wikipedia [14].
- Online 3D Virtual Labs: 3D rich graphical user interfaces will act as a powerful platform for the users to participate and perform collaborative activities, sharing results and exchanging media information among participants in a more natural way [15].
- 3D virtual shopping malls: where users can walk into different stores, see the products, interact with each other and walk into buildings.
- The Second Life community phenomenon**:** An example of the most popular 3D web application of Web 3.0 is Second Life [16].

C. Social Web

The web cannot be defined without connection to the human social realm as the Internet is a part of the technological infrastructure of society. The web is a social system in the sense of computer mediated cognition, communication and co-operation. Humans collectively generate an emergent informational structure on a system's scale. The integration of social web techniques (folksnomies, wikis) into the construction of so called "light weighted" community based Semantic Web solution, thus immediately integrating the social aspect of community with the technical level of the web to a new kind of intelligent symbiosis as the basic web layer [17].

In an article named Web 3.0: Basic Concepts, 2006 the author mentions that web 3.0 has already arrived and will be a social, collaborative movement such as Wikipedia, which has the human resources (in the form of the thousands of knowledgeable volunteers) to help create the ontologies (most likely as informal ontologies based on semantic annotations) that, when combined with inference rules for each domain of knowledge and the query structures for the particular schema, enable deductive reasoning at the machine level.

Technology advancements in Web 3.0 will take the current social computing to a new level called Semantic Social Computing or Socio-Semantic Web which will develop and utilize knowledge in all forms, e.g., content, models, services, & software behaviors [18].

D. Intelligent Web

The driving force for web 3.0 will be artificial intelligence. Semantic is defined as "meaning", that is web 3.0 will be designed to perform logical relationships between semantic meaning and information available in the Web. In other words, when searching for information in the web, web 3.0 will understand what the learner wants and suggests the information that suits the learner's needs depending on web 3.0 technology.

Traditional search engines can be frustrating to users. Users enter keywords for the search and then must evaluate typically sizeable results and determine which results are relevant. A semantic search engine utilizes semantics and knowledge coded into vocabulary sets which are interpreted by "smart agents" which then conduct intelligent searches returning pertinent information to the user [19].

As (Davies et al., 2005) states [20]: "Corporate knowledge workers need information defined by its meaning, not by text strings (''bags of words''). They also need information relevant to their interests and to their current context. They need to find not just documents, but sections and information entities within documents". In order to achieve this, the content of the web has to change through semantic technologies.

Wheeler [21] predicts that Web 3.0 will "not only promote learning that is more richly collaborative, it will also enable learners to come closer to 'anytime anyplace' learning and will provide intelligent solutions to web searching.

E. Media Centric

"Web 3.0 is media centric where users can locate the searched media in similar graphics and sound of other media formats. The pervasive nature of web 3.0 makes the users of web in wide range of area be reached not only in computers and cell phones but also through clothing, appliances, and automobiles." [13]

Web 3.0 searches will be able to find out the related similar media objects based on its features. The search engines would be able to take input(s) as a media or a multi-media object and will be able to search out related media objects based on its features. [12] For example, if a learner is searching about a certain animal, he can provide the search engine an input as an image of that animal and the search engine shall retrieve images of animals with similar features. The learner can also search with other media objects such as audio and video.

IV. Technologies of Web 3.0

The combination of semantic concepts with new technologies makes it possible to model data and capture the relationships between the data for machine learning. Semantic technologies tap new value by modeling knowledge, adding intelligence and enabling knowledge [22].

The ultimate goal of the Semantic Web is to transform the existing web into ". . . a set of connected applications . . . forming a consistent logical web of data . . ." [8] Put another way, it is the transformation of the Web from a linked document repository into a distributed knowledge base and application platform. This platform will allow for the vast range of content and services to be effectively exploited [9].

Semantic Web technologies allow us to build applications and solutions that were previously impossible and unfeasible. The combination of semantic concepts with new technologies makes it possible to model data and capture the relationships between the data for machine learning. Semantic technologies tap new value by modeling knowledge, adding intelligence and enabling knowledge [22].

Semantic Web technologies can be utilized across a variety of application areas. Data Integration involving the merging of data held in different formats across various repositories can provide better, domain specific search engine capabilities. The combination of these new semantic technologies, it is hoped will lead to the creation of intelligent software agents that will help facilitate knowledge sharing and exchange across the Web [23].

In order to gain a basic understanding of the Semantic Web, one must become familiar with the terminology:

URI

URIs or Uniform Resource Identifiers are used to identify resources; associating a URI to a resource makes the resource accessible and retrievable [24].

Metadata

Metadata is 'data about data'; a structured method for describing content so it can be easily accessed [19].

RDF

RDF stands for Resource Description Framework, the data model of the Semantic Web.

RDFS

The Resource Description Framework Schema extends the RDF standard with added functionality to specify domain vocabulary and object structures. It provides mechanisms for describing clusters of related resources and the relationships between these resources [25].

XML

XML or eXtensible Markup Language is a language that allows data to be shared on the Web and information to be exchanged between different platforms and applications [19].

RIF

RIF stands for Rules Interchange Format, a proposed component of the semantic web to be used in conjunction with OWL. It is a developing standard for exchanging rules among different systems, especially on the Semantic Web and is represented in the XML language.

SPARQL

Is the query language of RDF.

Ontologies

Ontologies define the terms used to describe and represent a specific area of knowledge or domain and are "essential building blocks" of the Semantic Web [26].

OWL

Web Ontology Language is the language used to create ontologies. OWL is built on RDFS and has the capability to express more complex and richer relationships [19].

Software agents

Software agents also referred to as smart agents, personal agents, or pedagogical agents, are "autonomous software entities, capable of performing specific tasks" [26].

Semantic markup

A semantic markup document is a file that describes the content of a Web page by using the terms defined in an ontology making the content of the Web page understandable by computers [19].

V. The Impact of web 3.0 on E-Learning

Web 3.0 and its implications for online learning are still evolving and a clear vision of "E-learning 3.0" is still in the future. Educators have the opportunity to influence emerging Web 3.0 technologies by helping to define that vision [27].

In web 3.0 the two major components are the learner and the 3D tutor. The learner is the human who intends to acquire knowledge about a specific subject. The tutor is an intelligent agent which delivers the collective knowledge to the learner [28].

A. The Impact of Web 3.0/Semantic Web-based Learning for Instructors

The Semantic Web has the means to assist instructors in course development, learner support, assessment, record keeping and document control tasks [29] [30]. The possibility of Web 3.0 will assist the instructor by creating reusable learning objects and providing immediate feedback to a learner at a specific stages of the learning process [31].

Research papers that include machine-readable markup will be more accessible to semantic search engines resulting in more precise returns to users' queries [32]. Clark, Parsia, and Hendler (2004) state "the added expressivity of the Semantic Web, coupled with search and query tools already under development, will allow changes in non-scientific fields as well. For example a number of historians could each annotate the same document to express differences of opinion about its comment, creating communities of deconstruction" [33]. Data collections (ontologies) from different fields will be linked creating "a network effect in academic knowledge" [33].

B. The Impact of Web 3.0/ Semantic Web-based Learning for Learners

Web 3.0 will offer personalized learning for learners. Learning modeling uses the learner's background knowledge, skills, aptitudes, motivations, learning and media preferences, mastery of content being taught, and learning progress to tailor the instruction to the learner [26] [29].

Personalized Learning can match the complexity of the instruction to the learners' needs. Devedzic notes that learner personalization "presents only the information that is really relevant for the learner, in the appropriate manner, and at the appropriate time" [26].

Smart agents will assist learners in documenting and archiving their learning products, locating resources, and working collaboratively [30].

VI. Definition of E-Learning Environment

E-Learning environment appears after learning environment impacted by network technology. As a new type of learning environment, it is causing people's popular attention. There is a various way of expressing it, such as web-learning environment, virtual learning, and E-learning environment digital learning environment and so on.

A good E-learning environment is made up of several functional modules. Professor Ke Qingchao in educational information technology college of South Normal University hold the views that the construction of E-learning environment should include independent learning tools module, collaborative learning platform modules, FAQ module, resources management module, and E-learning evaluation module five function modules.

VII. Tools of Web 3.0 for E-learning

While Education 1.0 was a one-way process, Education 2.0 uses the technologies of Web 2.0, Education 3.0 will create a much more free and open system focused on learning.

"Education 3.0 is characterized by rich, cross-institutional, cross-cultural educational opportunities within which the learners themselves play a key role as creators of knowledge artifacts that are shared, and where social networking and social benefits outside the immediate scope of activity play a strong role" [34].

Web 3.0 offers many tools and services for different kind of web applications on Internet:

A. 3D Wikis

With the evolution of 3D web, researchers & technocrats have been working on new projects to bring a new dimension to the world of Wikis & encyclopedia. Some examples of this kind of technology can be found on software like Copernicus-3D Wikipedia [35].

Suppose a Learner had performed the search and chose one of the results related to information about a specific geographical region, the camera will move to the particular place on the spinning globe to send relevant audio/video information.

For instance, the camera will "fly" towards the island of Ireland as a result of searching for Irish heritage park; eventually, the article about the Irish Heritage Park in Williamsburg will be presented to the user along with the video on Irish heritage park [36].

B. 3D Encyclopedias

A 3D Encyclopedia would be able to provide rich information involving all media and animation, for learners, so that they can have better impact on learning & knowledge.

C. 3D Virtual worlds

Virtual worlds can be seen as the beginning of new era of e-learning as they allow learners to interact in 3D Worlds for learning purposes for example to involve in virtual surgeries or to visit places that a learner can't visit in the real world like visiting planets, traveling inside the body of a human being, visiting various ancient places ..etc.

D. 3D Virtual Labs

These Web based Labs can prove to be quite beneficial for online learners . They could go to an immersive virtual science lab to do experiments. After the simulation, students could go offline into a real science lab to perform the correct experiment and see how it works [15].

E. Intelligent Search Engines

When you use a traditional Web search engine, the engine isn't able to really understand your search. It looks for Web pages that contain the keywords found in your search terms. Experts also believe that with Web 3.0, every user will have a unique internet profile based on that user's browsing history [13].

F. Online 3D Games

Can promote student collaboration where learners can come together, meet virtually, and collaborate together. They can fly over and move things around in a 3D world that is very much similar to their real world but with less cost and danger. In addition, Instructors and learners can introduce educational content into professionally designed 3D games.

G. 3D Avatars

Making a learner create a virtual 3D avatar, can encourage role playing. Learners can role play and become the profile we want to teach.

H. Intelligent Tutoring Systems

In this new era of e-learning, The learners have access the unbelievable knowledge source. The tutors are intelligent agents who are customized for the learners. Moreover, The learner can specify the tutor's avatar and gender.

Based on the intelligence and personal preferences of learner the tutor must deliver the knowledge. The tutor must collaborate knowledge from various web resources , filter the irrelevant knowledge and share it [28].

I. Synchronous/Asynchronous and Social Content

The content can be live (synchronous), meaning that when a learner interacts with the interface to add or change content, everyone is able to see the changes in real time without needing to refresh the page in the browser. The content can be asynchronous, meaning that any other learner who was off-line when a change took place, can review the changes any time he logs on the site. Furthermore, The web-site will contain on-line chat (both text and video) to enable communication between tutors and learners.

J. Wiki-enabled Interface

Interfaces of e-learning sites will be embedded with wikis. For example, tutors can define which words of the content will be linked to wikis.

VIII. ISSUES AND CHALLENGES

The web technologies developed over the last number of years to meet the needs of the Semantic Web are still in a premature phase. This, in turn, has delayed the implementation of the technologies and languages represented in the upper layers of the semantic stack. Standard solutions remain outstanding for these elements which include privacy, trust and proof.

As Anderson and Whitelock (2004) note [30], implementing the Semantic Web is more complicated than the current HTML-based, display-oriented Web 2.0. A lack of domain ontologies for learning courses is a barrier to the use of ontologies in e-learning systems and the ontology creation process is difficult and time consuming for instructors [37].

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